

POLYMORPHISM AND ABSTRACTION

This tutorial tests the basics of polymorphism and abstraction to be implemented using interfaces and classes (also includes basic inheritance).

Q. Today's smartphones execute a variety of utilities, such as cameras, cell phones and can connect to the internet using Wifi networks. You have to write code to create such a smartphone.

- 1) Firstly, this phone would have camera facilities, which can take a picture or record videos.
- 2) Secondly, the phone should be able to connect to a Wifi network. For this, two methods are required. One will be to get a list of all available networks, and the other method would require you to connect to any of these networks.
- 3) Also create a call phone. This phone can just make or receive calls, and has no other facilities. Each cell phone is identified by a Model Name, and by a 6 digit number that it hosts. This phone also has 2 methods- one takes another 6 digit number as input, and calls that number, and the other method just receives a call(do NOT take any arguments here).
- 4) Now create a smartphone to execute all the above facilities. Have a smartphone behave as a camera to record a video, and use another smartphone to get all Wifi network names, connect to a Wifi, and receive a call.

SOLUTION:

```
//package com.company;

public interface Camera{
    public void takeSnap();
    public void recordVideo();
}

public interface Wifi{
    public String[] getNetworks();
    public void connectToNetwork(String network);
}

public class CellPhone{
    private String modelName;
    private int phoneNumber;
    public CellPhone(String name, int number){
        this.modelName = name;
        this.phoneNumber = number;
    }
}
```

```

    public void callNumber(int phoneNumber){
        System.out.println("Calling "+ phoneNumber);
    }
    public void pickCall(){
        System.out.println("Connecting... ");
    }
}

public class SmartPhone extends CellPhone implements Wifi, Camera{
    @Override
    public void takeSnap(){
        System.out.println("Taking snap");
    }
    @Override
    public void recordVideo(){
        System.out.println("Recording Video");
    }
    @Override
    public String[] getNetworks(){
        System.out.println("Getting List of Networks");
        String[] networkList = {"Jio4G", "STUDENTS-N", "FACULTY-N"};
        return networkList;
    }
    @Override
    public void connectToNetwork(String network){
        System.out.println("Connecting to " + network);
    }
}

public class CellSystem { //This is the main class
    public static void main(String[] args) {
        Camera s1 = new SmartPhone(); // This is a smartphone but, use it as a camera
        // cam1.getNetworks(); --> Not allowed
        // cam1.sampleMeth(); --> Not allowed
        s1.recordVideo();
        SmartPhone s2 = new SmartPhone();
        String[] networks = s2.getNetworks();
        s2.connectToNetwork(networks[1]); // array is of size 3, so index 1 exists.
        s2.pickCall();
    }
}

```